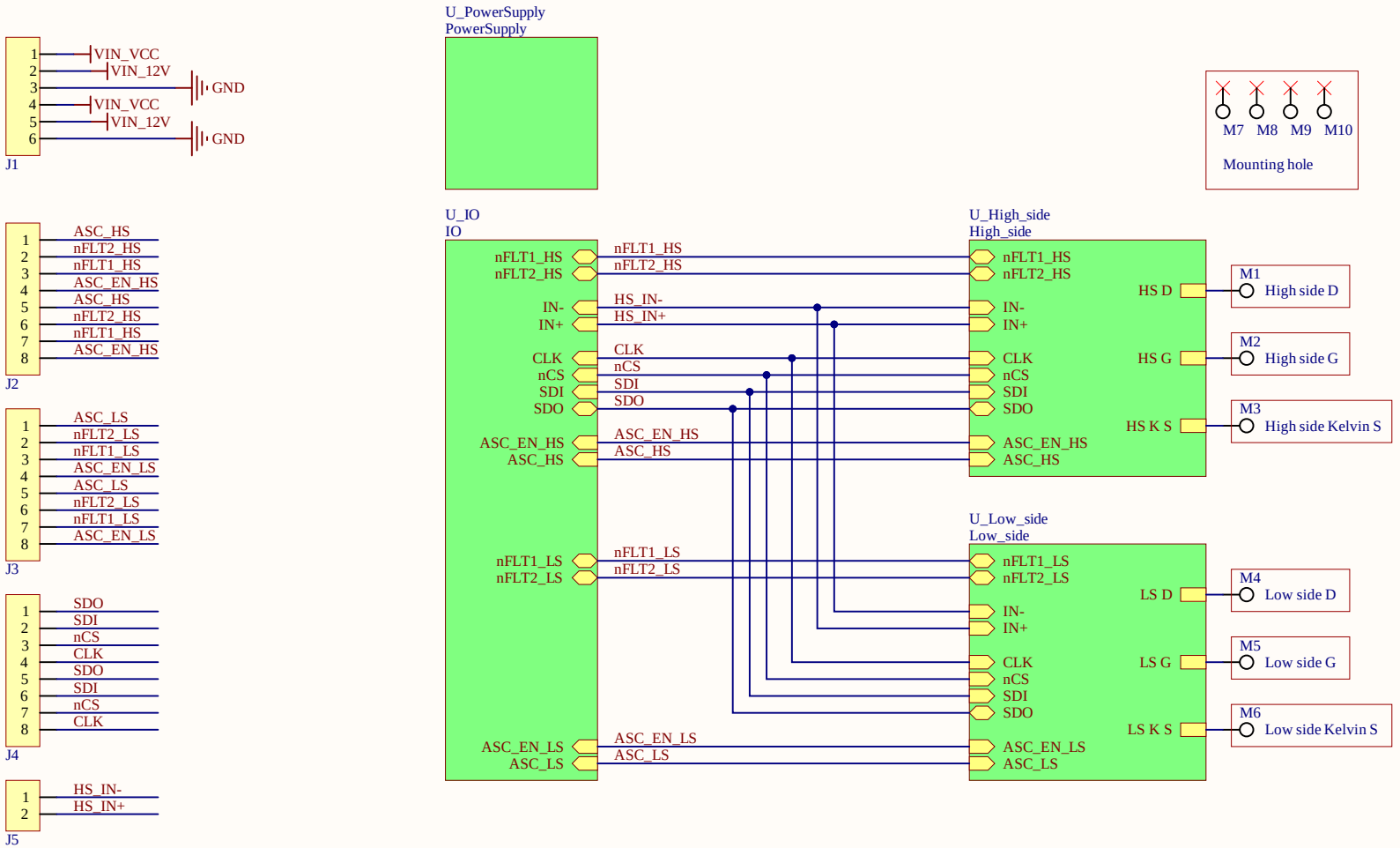
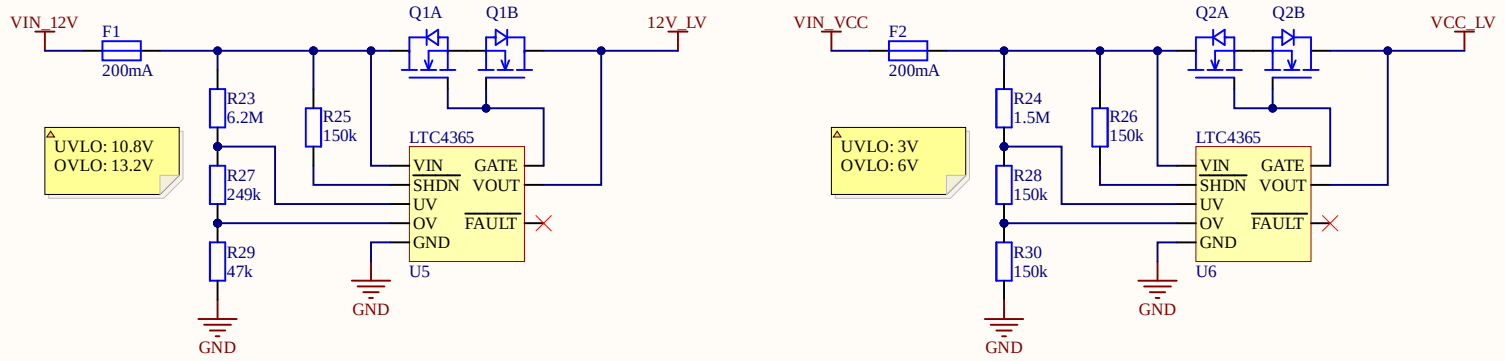
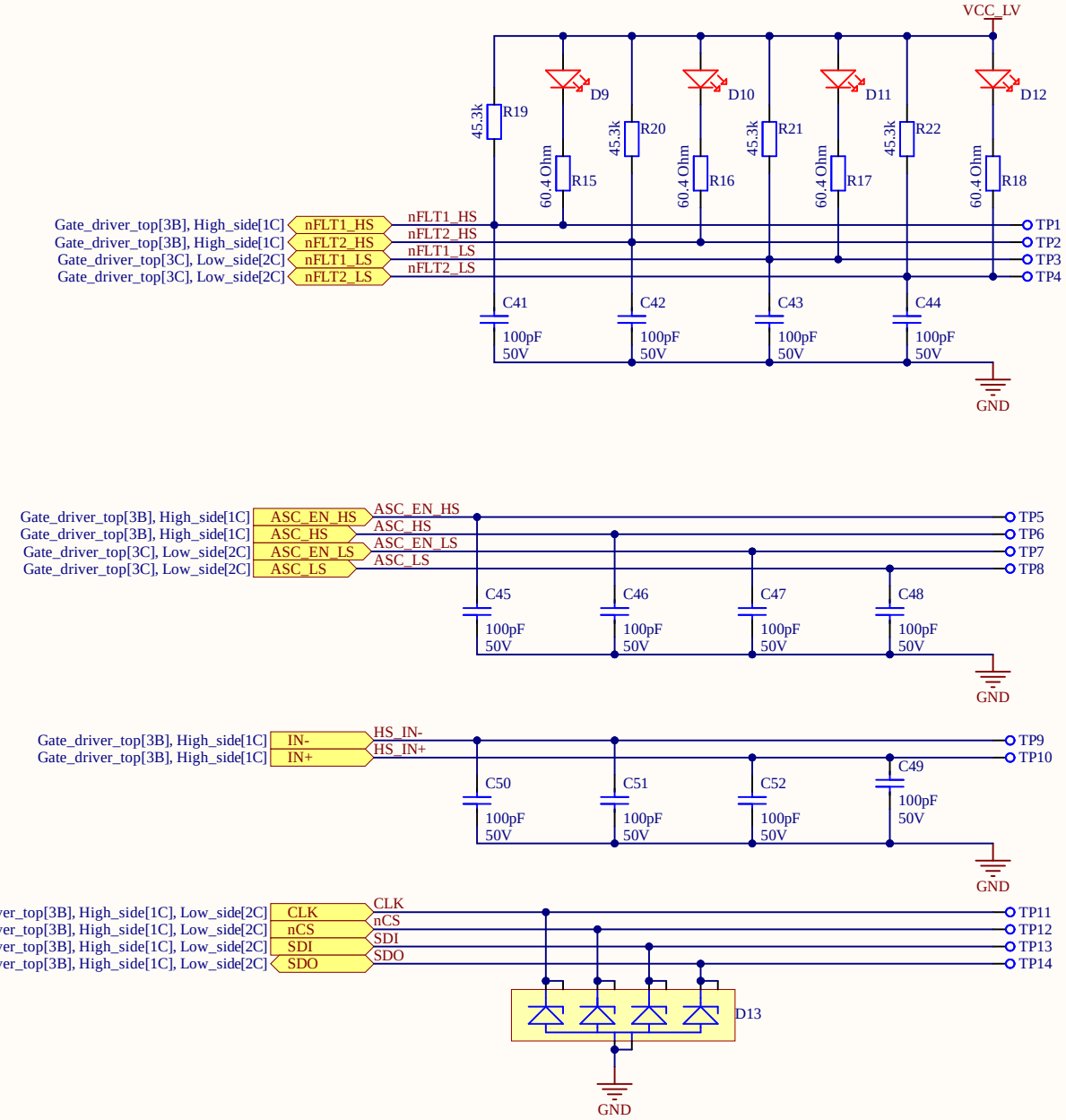


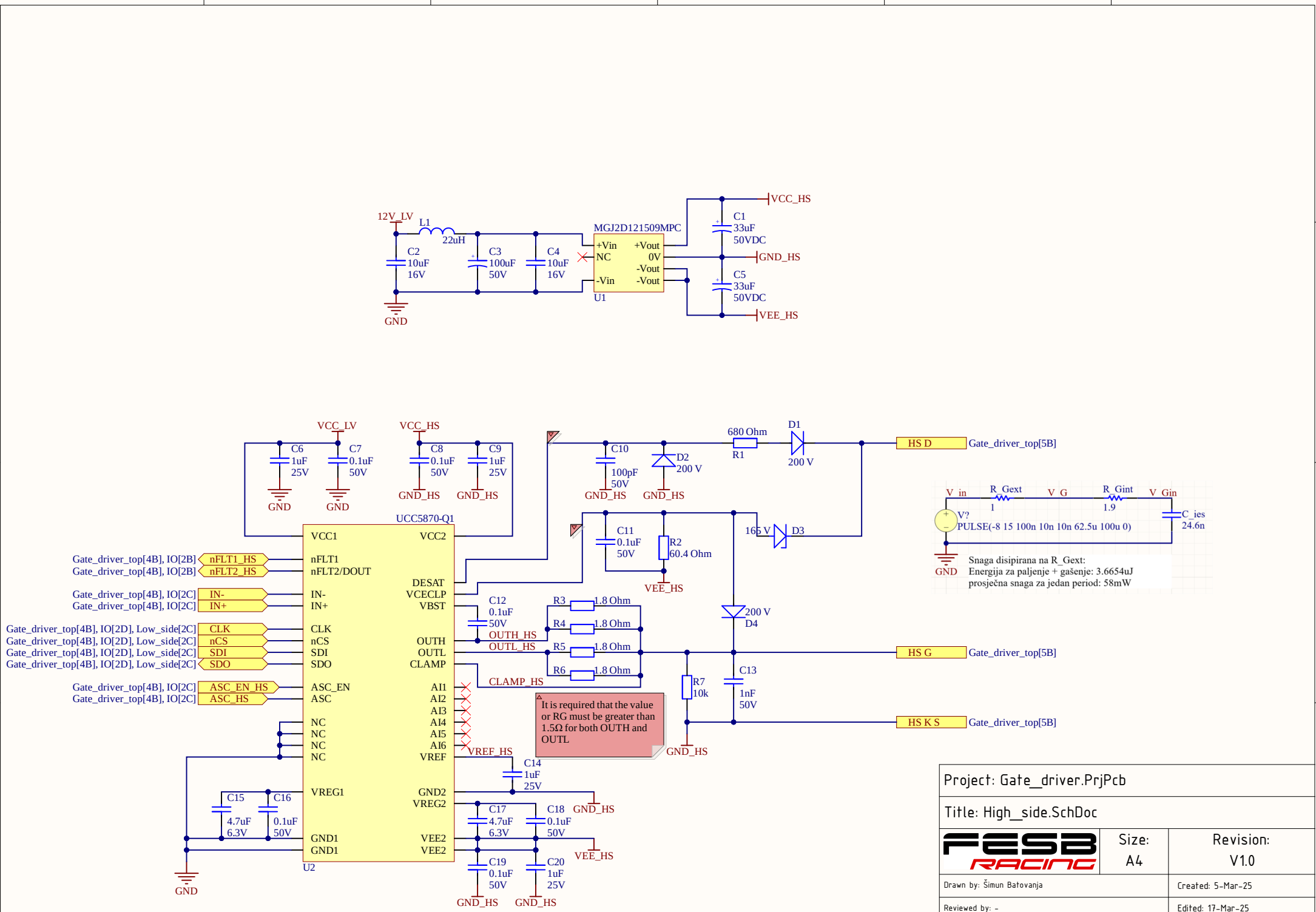
Project: Gate_driver.PrjPcb		
Title: Gate_driver_top.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 5-Mar-25
Reviewed by: -		Edited: 17-Mar-25

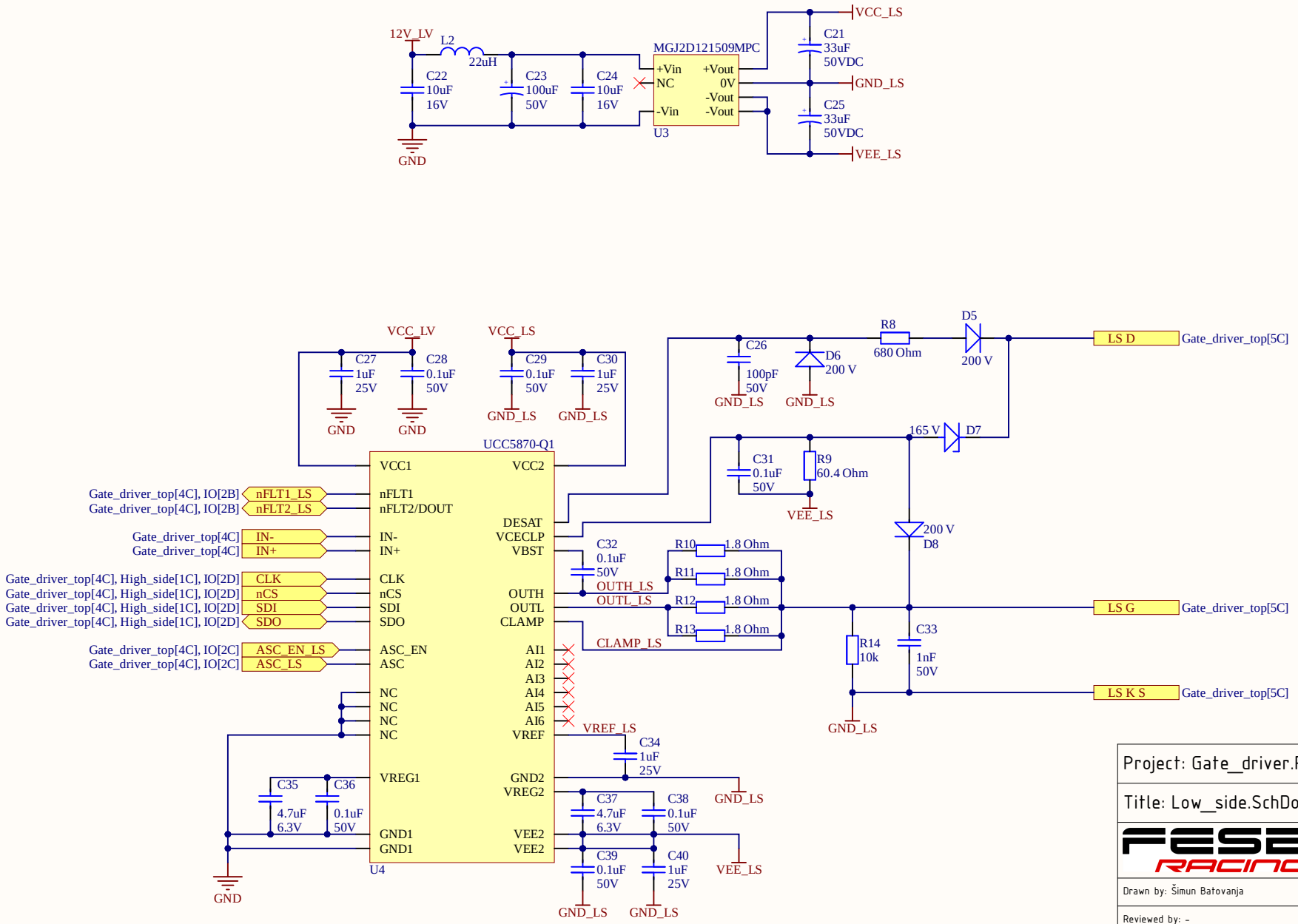


Project: Gate_driver.PrjPcb		
Title: PowerSupply.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 5-Mar-25
Reviewed by: -		Edited: 17-Mar-25



Project: Gate_driver.PrjPcb		
Title: IO.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 5-Mar-25
Reviewed by: -		Edited: 17-Mar-25


12V_LV
L1
22uH
C2
10uF
16V
C3
100uF
50V
C4
10uF
16V
GND
MGJ2D121509MPC
+Vin
NC
-Vin
+Vout
0V
-Vout
-Vout
U1
C1
33uF
50VDC
C5
33uF
50VDC
VCC_HS
GND_HS
VEE_HS
VCC_LV
VCC_HS
C6
1uF
25V
C7
0.1uF
50V
C8
0.1uF
50V
C9
1uF
25V
GND_HS
UCC5870-Q1
VCC1
VCC2
nFLT1
nFLT2/DOUT
DESAT
VCECLP
VBST
IN-
IN+
CLK
nCS
SDI
SDO
OUTH
OUTL
CLAMP
ASC_EN
ASC
AI1
AI2
AI3
AI4
AI5
AI6
VREF
NC
VREG1
VREG2
GND1
GND2
VEE2
Gate_driver_top[4B], IO[2B]
nFLT1_HS
nFLT2_HS
Gate_driver_top[4B], IO[2C]
IN-
IN+
Gate_driver_top[4B], IO[2D], Low_side[2C]
CLK
nCS
SDI
SDO
ASC_EN_HS
ASC_HS
OUTH_HS
OUTL_HS
CLAMP_HS
VREF_HS
C10
100pF
50V
D2
200 V
680 Ohm
R1
D1
200 V
HS D
Gate_driver_top[5B]
C11
0.1uF
50V
R2
60.4 Ohm
165 V
D3
C12
0.1uF
50V
R3
1.8 Ohm
R4
1.8 Ohm
R5
1.8 Ohm
R6
1.8 Ohm
200 V
D4
HS G
Gate_driver_top[5B]
R7
10k
C13
1nF
50V
HS K S
Gate_driver_top[5B]
It is required that the value or RG must be greater than 1.5Ω for both OUTH and OUTL
C14
1uF
25V
C15
4.7uF
6.3V
C16
0.1uF
50V
C17
4.7uF
6.3V
C18
0.1uF
50V
C19
0.1uF
50V
C20
1uF
25V
U2
V_in
R_Gext
1
V_G
R_Gint
1.9
V_Gin
V?
PULSE(-8 15 100n 10n 10n 62.5u 100u 0)
C_ies
24.6n
Snaga disipirana na R_Gext:
Energija za paljenje + gašenje: 3.6654uJ
prosječna snaga za jedan period: 58mW
Project: Gate_driver.PrjPcb
Title: High_side.SchDoc
Size:
A4
Revision:
V1.0
Drawn by: Šimun Batovanja
Created: 5-Mar-25
Reviewed by: -
Edited: 17-Mar-25



Project: Gate_driver.PrjPcb	
Title: Low_side.SchDoc	
FESB RACING	Size: A4 / Revision: V1.0
Drawn by: Šimun Batovanja	Created: 5-Mar-25
Reviewed by: -	Edited: 17-Mar-25